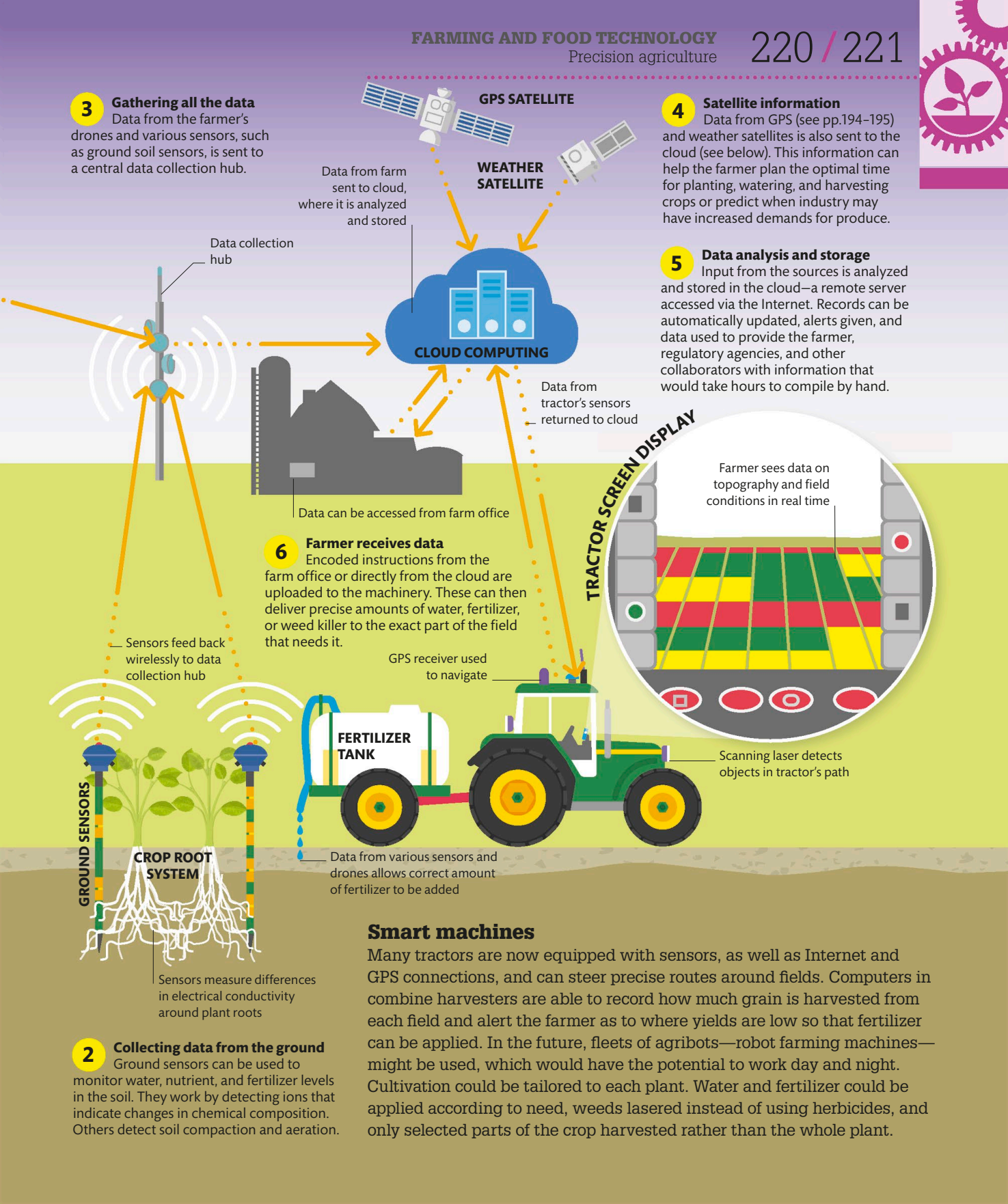


3 Gathering all the data
Data from the farmer's drones and various sensors, such as ground soil sensors, is sent to a central data collection hub.

4 Satellite information
Data from GPS (see pp.194–195) and weather satellites is also sent to the cloud (see below). This information can help the farmer plan the optimal time for planting, watering, and harvesting crops or predict when industry may have increased demands for produce.

5 Data analysis and storage
Input from the sources is analyzed and stored in the cloud—a remote server accessed via the Internet. Records can be automatically updated, alerts given, and data used to provide the farmer, regulatory agencies, and other collaborators with information that would take hours to compile by hand.

6 Farmer receives data
Encoded instructions from the farm office or directly from the cloud are uploaded to the machinery. These can then deliver precise amounts of water, fertilizer, or weed killer to the exact part of the field that needs it.

2 Collecting data from the ground
Ground sensors can be used to monitor water, nutrient, and fertilizer levels in the soil. They work by detecting ions that indicate changes in chemical composition. Others detect soil compaction and aeration.

Smart machines

Many tractors are now equipped with sensors, as well as Internet and GPS connections, and can steer precise routes around fields. Computers in combine harvesters are able to record how much grain is harvested from each field and alert the farmer as to where yields are low so that fertilizer can be applied. In the future, fleets of agribots—robot farming machines—might be used, which would have the potential to work day and night. Cultivation could be tailored to each plant. Water and fertilizer could be applied according to need, weeds lasered instead of using herbicides, and only selected parts of the crop harvested rather than the whole plant.